

- 4 (a) A resistor of resistance of $8.4 \text{ k}\Omega$ and a LDR (light dependent resistor) are connected in series, as shown in Fig. 4.1. The resistance of the LDR when light is shone on it is $400 \text{ }\Omega$.

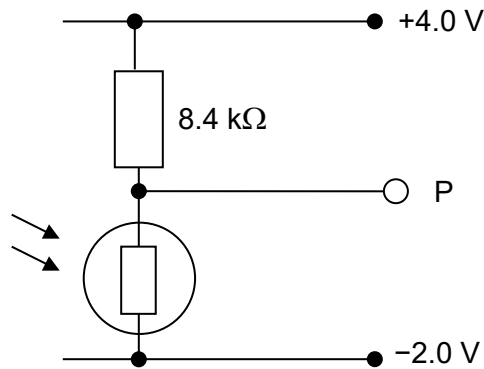


Fig. 4.1

- (i) Calculate the potential at point P when light is shone on the LDR.

potential = V [2]

- (ii) An alarm is now connected in parallel with the LDR in Fig. 4.1 and a laser beam is directed continuously at the LDR in a simple burglar alarm system. When a burglar blocks the laser that is shining on the LDR, the alarm is triggered. Assume that the resistance of the alarm is very large.

Explain how the LDR is used to trigger the alarm in such a system.

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 [2]

- (b) A circuit is set up as shown in Fig. 4.2. The potentiometer wire XY has a resistance of $1.6\ \Omega$ and is 1.00 m long. The movable contact J can be connected to any point along wire XY.

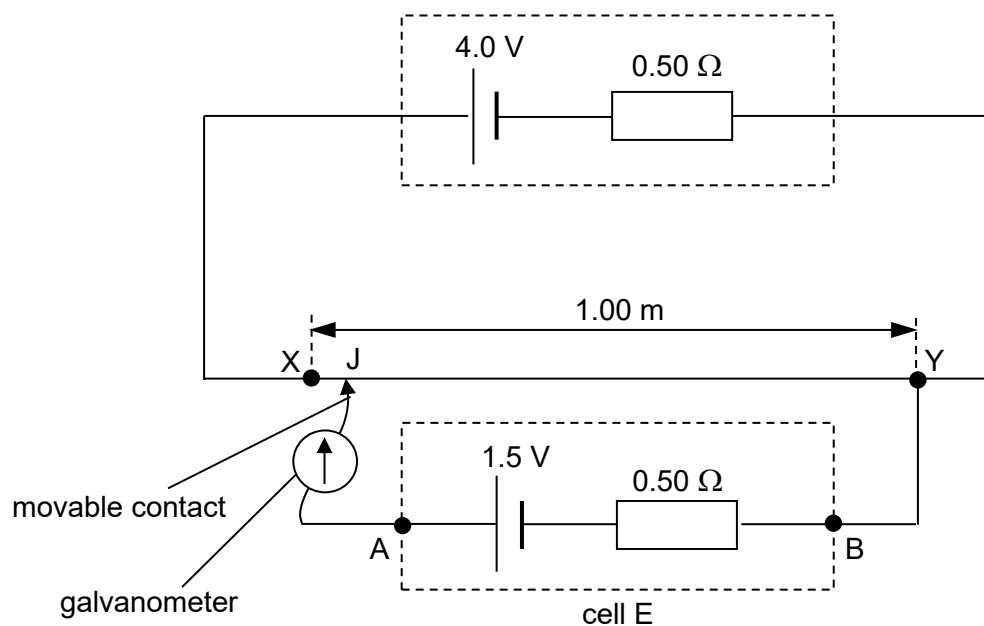


Fig. 4.2

- (i) Determine the distance of the contact J from Y, such that there is no deflection in the galvanometer.

distance = m [2]

(ii) A $1.5\ \Omega$ resistor is then connected across points A and B.

Determine the distance of the contact J from Y, such that there is no deflection in the galvanometer.

distance = m [3]

5 Fig. 5.1 shows an electron passing an earthed vertical plate P with negligible speed and